Question			Answer	Marks	AO	Guidance
9	(a)		$\frac{dy}{dx} = e^x - 4e^{-x} = 3$	M1	2.1	Equate to 3
			$(e^x)^2 - 3e^x - 4 = 0$	E1	2.1	AG Rearrange to quadratic in e^x Also allow $e^{2x} - 3e^x - 4 = 0$ Expression $= 0$ must be seen
				[2]		
9	(b)		Solve to give $e^x = [-1], 4$	M1	2.1	May be BC giving at least one root of the quadratic equation (a)
			When $e^x = 4, x = \ln 4$	A1	2.1	$x = \ln 4$ must be seen explicitly
			When $e^x = -1$ there are no real values of x, so no other points on the curve.	E1	2.1	must explain why they reject the value –1 for e^x, or state $e^x + 1$ is never zero
				[3]		
9	(c)		Equation $y = \int (e^x - 4e^{-x})\,dx$			
			$[y =] e^x + 4e^{-x} [+c]$	B1	3.1a	Condone missing $+c$ in their integral
			When $x = 0, y = 0 = 1 + 4 + c$	M1	3.1a	Attempt to evaluate c
			So $c = -5$ $[y = e^x + 4e^{-x} - 5]$	A1	1.1b	
			When $x = 1, y = e^1 + 4e^{-1} - 5$	M1	2.1	Substituting $x = 1$ into their expression
			$y = -0.810 < 0$ so below the x-axis	E1	2.1	AG must argue below the axis from correct y value. Must be clear that -0.81 is a y-coordinate
				[5]		